Controlled Packet Capture and 802.11 Protocol Analysis Lab - Lab Setup-aircrack-wireshark

Great! We have our lab setup! Now what? That is a great question.

In this post, we will explore the Aircrack-ng suite, its individual components, and how they can be used for wireless network monitoring and packet capture.

Next, we will dive into Wireshark setup. We will cover how to use monitor-mode interfaces to capture wireless frames, apply display filters to quickly identify relevant traffic, and import community-developed display filters to enhance our analysis workflow.

Aircrack-ng

Let's get started!

What Is Aircrack-ng

Aircrack-ng is an open-source wireless network security auditing suite used to evaluate Wi-Fi security. It provides tools for monitoring wireless networks, capturing packets, analyzing wireless traffic, and testing wireless encryption.

Aircrack-ng Tools

- airmon-ng - Manages wireless interfaces and monitor mode.
- airodump-ng - Captures and analyzes wireless network traffic.
- aireplay-ng - Supports wireless packet injection during authorized testing.
- aircrack-ng - Analyzes captured data to assess wireless encryption security.

Throughout this series, we will rely primarily on "airmon-ng" to assist with creating monitor interfaces that will later be utilized for packet capture, and "airodump-ng" to analyze network traffic. This series will not engage with "aireplay-ng" or "aircrack-ng" as both tools are outside of the intention of this series.

Monitor Interface

What is a monitor interface?

An interface placed in monitor mode allows a wireless network adapter to capture 802.11 wireless frames, including traffic not specifically addressed to the adapter. Let's identify if our adapters support monitor mode.

Verify adapters support monitor mode

Recall our 'iw' utility from the lab setup?

```
iw phy
```

Review the output. Specifically, "Supported interface modes."

To minimize the output, you could also issue.

```
iw phy | grep -E -i -A 10 "interface modes|phy"
```

Result

```
Supported interface modes:
         * IBSS
         * managed
         * AP
         * AP/VLAN
         * monitor
         * P2P-client
         * P2P-GO
         * P2P-device
```

Great! We have confirmed our adapters support 'monitor mode'.

Create a monitor interface

To create a monitor interface, will utilize airmon-ng. The process is simple: we pass the wlan adapter name to airmon-ng. Airmon-ng then creates a monitor interface. This new interface will be named 'wlan0mon' and will be used by Wireshark as a capture interface.

Create Interface

Side note: notice the 'sudo'? We will need to elevate privileges to create a new interface.

```
sudo airmon-ng start wlx9cefd5f6950a
```

Result

Output from airmon-ng showing a newly created monitor interface named "wlan0mon"

```
(mac80211 monitor mode vif enabled on [phy2]wlan0mon)
```

Verify

Use the 'iwconfig' command.

```
iwconfig
```

New monitor interface wlan0mon

Notice the "Mode"?

```
wlan0mon  IEEE 802.11  Mode:Monitor
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Power Management:on
```

Excellent! We have successfully created a monitor interface to use with Wireshark.

Wireshark

We installed Wireshark in the previous Lab Setup post. Now it's time to get Wireshark fired up and add a few options that will make our job much easier.

First, let's clear up a common stumbling block. During the installation of Wireshark, you were asked if you would like to give everyone the ability to use Wireshark. The reasoning is that Wireshark has to have access to the physical layer to capture. This type of access on Linux systems requires 'sudo' or elevated privileges.

If you did not select yes to allow users other than root access, no worries. There is a minor adjustment. During installation there was a group created called 'wireshark'. All users placed in this group will have the required privileges. So, if you are getting a warning stating you don't have the required privileges, all you have to do is place your user in the mentioned 'wireshark' group.

Add user to wireshark group

Use `$USER` instead of manually specifying the username. This is a Linux variable that means "the current logged in user". The Linux command 'usermod' is used to modify user permissions. In this case, add "-a" to group 'G'.

```
sudo usermod -aG wireshark $USER
```

Reconfigure permissions

```
sudo dpkg-reconfigure wireshark-common
```

Select Yes, then log out and back in.

Verify

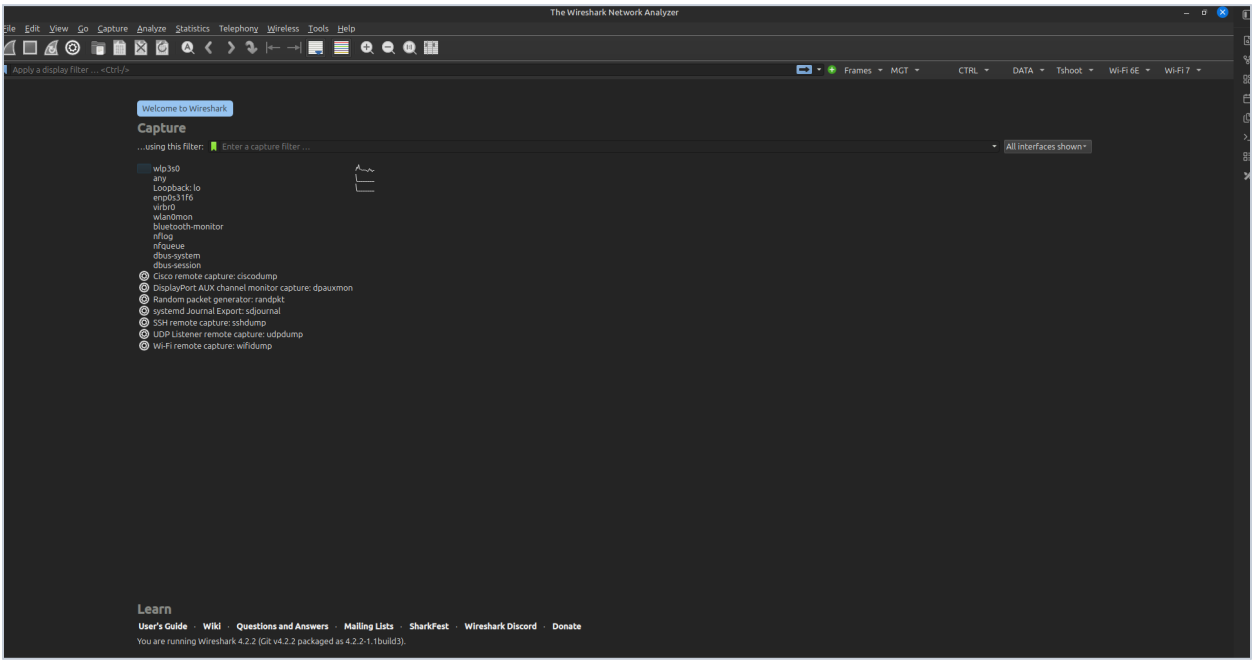
```
groups $USER
```

You should see `wireshark` in the group list.

Initialize Wireshark

Okay, all set! Let's fire up Wireshark. Click the Wireshark Icon.

Notice the 'wlan0mon' interface that we created with airmon-ng?



Fantastic! We are all set! Before we dive in, let's make our lives a bit easier! How do we do that? Through collaboration!

Community

If you have been in, near, or around the wireless community, you have or will quickly realize that at the foundation is a core concept. Community! I have been a part of many professional communities over my career and I can say, without a doubt, the wireless community is fantastic! Learn and share! Grow together. That is it. It's not complicated!

To that end, and in the spirit of community, the following are references to community-provided "wireshark" profiles. The profiles are the result of a lot of hard work by a lot of people. I am grateful for the effort and the kindness to share. I will continue to spread the good word and praises!

We will download, install, and explore throughout the series.

Wireshark Profile

WLAN Professionals - I encourage anyone in the wireless engineering field to follow WLAN Professionals. These folks are amazing and the level of knowledge you won't find anywhere else.

Download Profile Zip
https://wlanprofessionals.com/wlan-pros-master-profile-for-wireshark/

Rasika Nayanajith - I have been following this gentleman for years. Fantastic resource for all things wireless. https://mrncciew.com/

Download Profile Zip https://mrncciew.com/2025/09/02/get-rockstarwifi-wireshark-profile/

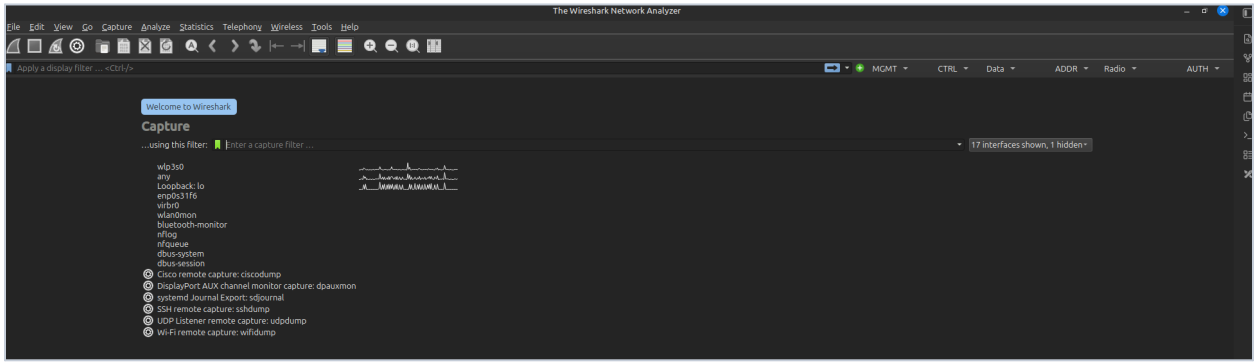
Install Profile

At the bottom-right corner of Wireshark, you will see 'Profile'. Right-click, select import, then select 'From zip file'. Repeat this process for each profile you want to add.

Result

If we look at our 'Filter Toolbar' we can now see multiple groupings of display filters.

!! Please explore !!



Conclusion

We have explored the aircrack-ng suite, its various tools, and how to utilize airmon-ng to create a monitor interface that we will use via Wireshark for packet capture. We set up a base configuration for Wireshark which included importing community-provided Wireshark profiles. The Wireshark profiles provide various display filter options that will assist with our packet analysis efforts. In our next post in this series we will start capturing packets and looking at the structures of wireless frames.